

Functional Requirements

1. User Authentication & Session Management

- The system shall allow users to log in using their email and password.
- The system shall validate user credentials against a secure database.
- The system shall create and manage a secure session token upon successful authentication.
- The system shall redirect users to a role-specific dashboard (User, Lawyer, Admin) after login.
- The system shall handle login failures (invalid credentials, locked accounts) with appropriate error messages.

2. Role-Based Registration

- The system shall allow new users to register by selecting a role (User/Client or Lawyer).
- The system shall present a dynamic registration form based on the selected role.
- For Lawyers, the system shall collect and validate professional license information and initiate an admin verification workflow.
- The system shall send an email verification link to complete the account creation process.
- Upon verification, the system shall create a user profile and send a welcome email.

3. AI-Powered Document Analysis

- The system shall allow authenticated users to upload legal documents in supported formats (e.g., PDF, DOCX).

- The system shall validate file type and size.
- The system shall queue uploaded documents for AI processing.
- The AI module shall extract text, identify key clauses, and assess potential risks within the document.
- The system shall generate and display a structured analysis report with insights and lawyer suggestions to the user.
- The system shall save the analysis report to the user's history.

4. Intelligent Lawyer Search s Recommendation

- The system shall allow users to search for lawyers based on criteria such as specialization, location, and ratings.
- The system shall filter and rank lawyers using an AI algorithm based on relevance to the user's case and search criteria.
- The system shall display lawyer profiles, including availability calendars and contact/booking options.
- The system shall allow users to directly select a lawyer for consultation or booking.

5. Appointment Booking s Management

- The system shall allow users to view a selected lawyer's available time slots.
- The system shall allow users to book, reschedule, or cancel appointments.
- The system shall require mutual approval (User C Lawyer) for any rescheduling changes.
- The system shall allow lawyers to manage their availability by adding or removing time slots.
- The system shall send automatic notifications for all appointment-related events (booking, rescheduling, cancellation).

6. AI Contract Template Generation

- The system shall provide a library of legal contract templates for various purposes.
- The system shall guide users through a form to input specific details required for the selected contract type.
- The AI shall dynamically customize the contract template based on user inputs.
- The system shall generate a final contract document (e.g., PDF, DOCX) for the user to download, review, and edit.
- The system shall save generated contracts to the user's history.

7. IR Smart Legal Search

- The system shall allow users to perform searches using natural language queries.
- The system shall process queries using NLP to understand intent and key concepts.
- The system shall employ a dual search strategy (keyword-based and semantic/vector-based) across legal documents and articles.
- The system shall combine, rank, and present search results by relevance.
- The system shall save user search history.

8. Admin - Lawyer Management

- The system shall provide an admin dashboard to view and manage lawyer accounts.
- Admins shall be able to verify lawyer credentials (license review) and approve/reject registrations.
- Admins shall be able to view, edit, and deactivate lawyer profiles.
- All admin actions on lawyer accounts shall be logged in the AdminLog.

9. AI Contract Analysis

- The system shall allow users to upload contracts for in-depth AI analysis.
- The AI shall run parallel analyses for clause identification, risk detection, and compliance checking.
- The system shall compile the results into a comprehensive report highlighting risks, non-standard clauses, and suggestions.
- The report shall include an option for the user to seek a lawyer consultation based on the findings.

10. Admin - Log Management

- The system shall maintain a secure audit log (AdminLog) of all significant system activities and user/admin actions.
- Admins shall be able to access, view, filter, and search through the system logs.
- Admins shall be able to export log data and generate summary reports.
- All access and actions within the log management interface shall themselves be logged.

Non-Functional Requirements

1. Performance

- The system shall support concurrent access by at least 10,000 users.
- Page load times for the dashboard shall be under 3 seconds under normal load.
- AI analysis tasks (document/contract) shall provide initial results within 60 seconds for standard documents (<50 pages).

2. Security s Privacy

- All user credentials and personal data shall be encrypted in transit (TLS 1.2+) and at rest.
- The system shall comply with relevant data protection regulations (e.g., GDPR).

- Strict role-based access control (RBAC) shall be enforced to ensure users can only access authorized functions and data.
- All authentication attempts and sensitive actions (logins, profile edits, document uploads) shall be logged for security auditing.

3. Usability s Accessibility

- The system shall have an intuitive user interface designed for users with varying levels of technical and legal expertise.
- The platform shall be available in at least two languages (English C Arabic) as per the project scope.
- The web interface shall be responsive and function correctly on major desktop and mobile browsers.

4. Reliability s Availability

- The core platform shall have an uptime of 99.5%.
- The system shall implement data backup and recovery procedures to prevent data loss.
- Critical services (like authentication and appointment booking) shall be designed with fallback mechanisms to handle partial failures gracefully.

5. Scalability s Maintainability

- The system architecture shall be modular to allow independent scaling of components (e.g., web server, AI processing queue, database).
- The codebase shall be well-documented to facilitate future maintenance and feature addition by new developers.

6. Legal s Compliance

- The system shall include clear disclaimers stating that its AI analysis and generated templates are for informational purposes only and do not constitute legal advice.
- The platform shall not offer services that constitute unauthorized practice of law (e.g., court representation).